

Technical Portfolio

Lillian Hunt



ABOUT ME

01 / EDUCATION

Junior studying Aerospace Engineering with a minor in Computer Science at the University of Illinois at Urbana-Champaign.

02 / INDUSTRY EXPERIENCE

Former Design Engineering Intern at LibertyWorks Rolls-Royce, focusing on mechanical design for advanced R&D contracts, specifically in flow meter design for high-temperature and high-pressure systems.

03 / CORE COMPETENCIES

Early-phase component design, 3D modeling, FEA verification and system definition.

04 / CAREER FOCUS

Passionate about early-phase design for deep space exploration, human-centered astronautics, and propulsion systems.



Design and Manufacturing Lead (2025-2026)

Conceptualized mechanical design for adjustable tool cart for a NASA challenge



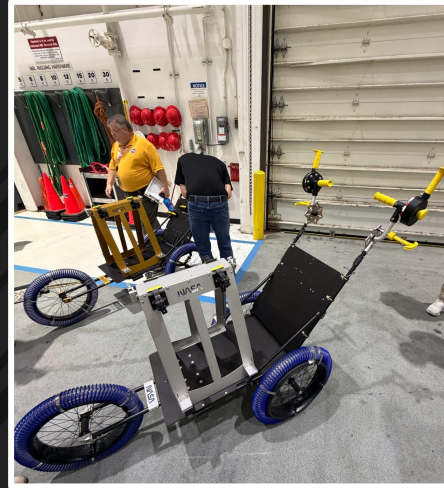
Propulsion R&D Intern (Summer 2026)

Matured early phase mechanical component design (pictured in an RB-211)

ATHENA: Adjustable Tool-cart Handle for Efficient Navigational Assistance

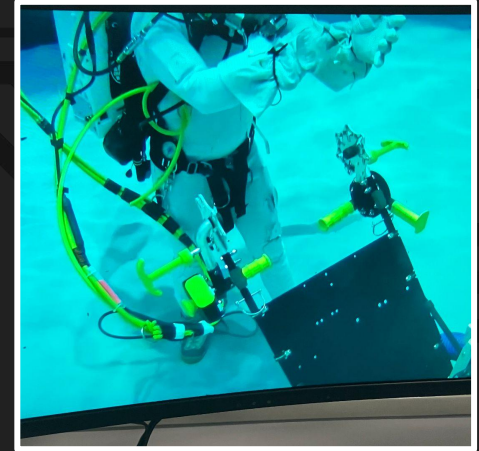
Served as the Design and Manufacturing Lead (2025-2026) for the UIUC Micro-g NExT team, a yearly NASA competition, developing the ATHENA adjustable tool cart handle for Artemis III lunar surface navigation.

- Designed dual independent adjustment mechanisms accommodating the 1st to 99th percentile of astronauts, exceeding NASA's set height range of 35"-50".
- Led team in generating all CAD deliverables as well as physical hardware, including components, assemblies, technical drawings, animations, FEMS, and all prototypes.
- Successfully tested the finalized design at NASA Johnson Space Center's Neutral Buoyancy Lab, performing above expectations – being apart of the 30% of teams who's tool survived functional testing at the NBL.



Tool Setup

ATHENA integrated onto the Tool Cart featuring dual independent adjustment mechanisms fabricated from 6061-T6 aluminum.



Tool Demonstration

Divers validating the tool design under simulated microgravity conditions at NASA's Neutral Buoyancy Lab.

FaNG: Fabric Anchoring Gadget

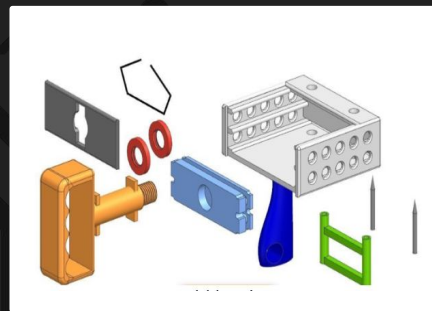
Served as the Research and Testing Lead (2024-2025) for the UIUC Micro-g NExT team, developing FaNG for ongoing repairs aboard the International Space Station.

- Designed independent mechanisms for creating pilot holes in MLI fabric and inserting an elastic clip to bind fabric together while minimizing pinch points, human interaction with the interface, and astronaut risk.
- Led material research and developed cost analysis trade studies, balancing prototyping costs with team logistics and travel budgets
- Recorded and performed analysis on the results of prototype testing — writing up reports that would inform any changes made to the tool design and the development of the next iteration.



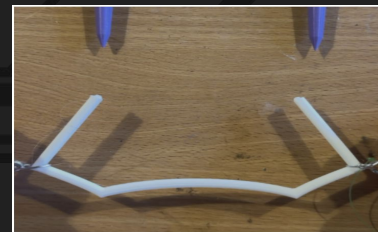
Prototype

Posing with FaNG prototype after placing 3/60 at the 2025 AIAA region III student conference



Rendering - Exploded View

CAD Rendering of the FaNG tool using Siemens NX



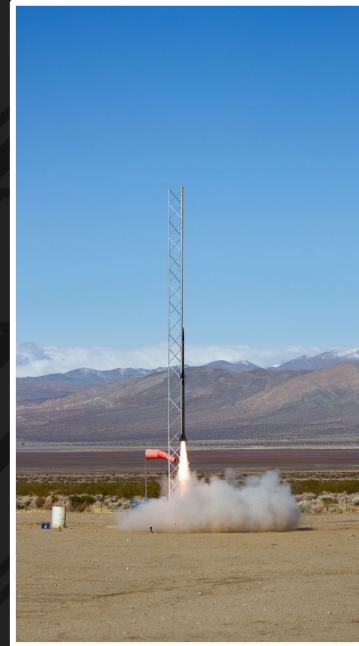
Movement Testing

Examined material performance across design iterations of clips to ensure they flexed the distance of the pilot holes without damage across multiple trials

Aether 2 Structural Redesign

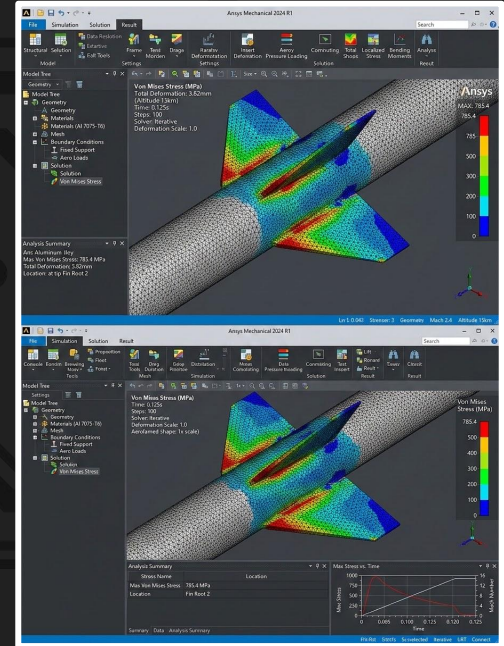
Aided the development of a high powered rocket as a general member (2024-2025) of the Structures, Design, and Analysis subteam, by contributing to the design of the rocket through verification, simulations, and hand calculations.

- Redesigned fins for Aether II, increasing booster fin span to 5.75" and sustainer fins to 3.5", to improve dynamic stability and multi-layer carbon fiber fabrication reliability after adhesion failure on Aether I.
- Verified design with FEA simulation, ensuring the design would survive expected loading conditions at max q
- Redesign led to a successful flight, helping Aether II set a new university altitude record with 75,124 feet.
- Wrote a custom script to parse avionics data, discovering that a crooked launch rod caused an approximate 20,000-foot loss in altitude.



Launch

Aether 1 launching. ~12 seconds prior to RUD



Simulation

Ansys FEA simulation validating structural integrity.

Cassie Avionic Systems

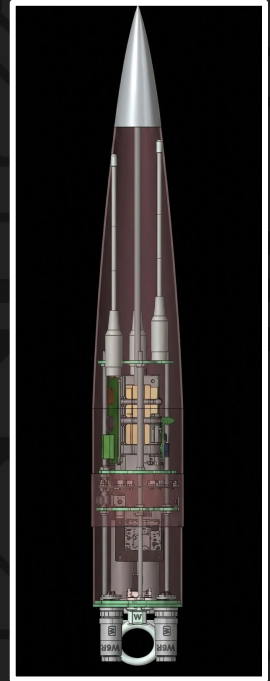
Supported high-powered rocketry team as a general member (2025-2026) of AV-structures subteam, aiding in the development and construction of AV systems flown throughout the year

- Contributed significant portions of the CAD design for the L3 class rocket Cassie's Avionics optimizing for a limited overall size and form factor
- Designed new Recovery + AV bulkhead, integrating cases for second stage ejection charges to conserve 2" of space in Cassie's body tube.
- Decreased the form factor of the battery stack, allowing critical subcomponents to remain on the bulkhead while adhering to the desire to decrease the size of the AV Bay.



Preparing for Launch

Cassie vertical, being prepared to launch. Argonia, Kansas



Rendering

Full AV Bay CAD after partial redesign

MonARCH CubeSat: Thermal Analysis & Simulation

Developed transient thermal models for the Laboratory for Advanced Space Systems (LASSI) at Illinois to validate satellite survivability.

- Built transient thermal simulations using Ansys Thermal Desktop and SINDA to define bounding hot and cold orbital cases at a 400 km altitude.
- Established internal power dissipation models tied to mission modes and implemented environmental heating using radiative coupling and view factors.
- Proved passive thermal control was insufficient for systems like avionics, establishing a credible baseline for the final 100+ node thermal model.



Render

Conceptual Render of MonARCH cubesat in LEO

SPECTRA: Solar-illuminated Precision Emissions CubeSat Tracking Radiative Absorption

As part of AE298, SPECTRA was created as a satellite proposal. SPECTRA is a cubesat designed to track atmospheric radiative absorption of CO₂. This enables the ability to pinpoint locations to allocate more resources to fight global warming.

- Developed the CubeSat's mission concept and technical design, including block diagram CAD models, aligned with principal investigator (PI) requirements
- Authored the Mission Concept Review (MCR), demonstrating technical feasibility and securing simulated funding in a competitive small-sat lab review.
- Managed mission budgeting within a \$1.5M conceptual grant, ensuring alignment of system design with cost and mission objectives.

